V2X-Sim — Sprint 5 Report

Collision Realism & Vehicle-to-Vehicle (V2V) Communication

Cooperative Perception at Smart Intersections
A CARLA study of 5G NR-V2X with ETSI CPM Release 2 under realistic edge-compute and latency constraints in mixed traffic

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Executive summary

Sprint 5 hardened the simulation's realism and added vehicle-to-vehicle communication on top of the existing roadside-to-vehicle (V2I) stack. Four correctness fixes (collision counting, driver-role behaviour, spawn overlap, hostility scope) plus a new V2V layer turned the ablation from a metric with inflated, hard-to-trust numbers into a clean, reproducible safety experiment. Two headline results emerged: (1) **vehicle penetration is the dominant safety lever and it generalises across maps** — replacing hostile human drivers with careful connected-autonomous vehicles collapses collisions monotonically; and (2) **V2V's marginal value over roadside-only perception depends on scenario difficulty** — neutral on the easy in-distribution map, but a clear collision reduction on the harder held-out map. The test suite grew 215 -> 241, all passing.

1. Scenario and project context

The project studies **cooperative perception at signalised intersections**: roadside units (RSUs) with cameras detect road users and broadcast what they see to connected vehicles, which fuse those messages with their own sensing to act earlier and more safely in mixed traffic. The simulator is built on CARLA 0.9.16 and models the full pipeline end to end rather than assuming an idealised channel.

Pipeline. RSU camera -> YOLO26 detector -> ETSI CPM message -> V2X channel (packet loss + latency + channel-busy ratio) -> connected-vehicle late-fusion -> braking decision.

Actors / roles.

- **HDV** — human-driven vehicle, modelled by CARLA's Traffic Manager with behavioural driver profiles (attentive / distracted / aggressive). The source of conflicts.
- **CAV** — connected autonomous vehicle: a Traffic-Manager vehicle that additionally runs the cooperative-perception fusion and braking logic, and (Sprint 5) the V2V layer.
- **RSU** — roadside unit: one overhead camera at an intersection running the detector and broadcasting ETSI CPMs.

Penetration rate is the fraction of vehicles that are CAVs: p=0.0 is an all-HDV baseline, p=0.5 is mixed traffic, p=1.0 is a fully connected fleet. It is the primary independent variable of the ablation.

2. Where we were: Sprints 1–4

Sprint 5 builds directly on four prior sprints. The table summarises what each delivered.

Sprint	Delivered	Key outcome
1 — RSU infrastructure	Deterministic intersection discovery; single-camera RSU (worked around a CARLA 0.9.16 multi-camera crash).	Programmatic RSUs on Town03/04/05/10.
2 — Detector	YOLO26 fine-tuned on Town03+04+05 (4 classes: vehicle / motorcycle / bicycle / pedestrian).	Held-out Town10: mAP@50 = 0.53, precision 0.85, recall 0.50.
3 — V2X protocol stack	CARLA-agnostic: ETSI CPM encoder, Coll-Perales latency model, Thandavarayan packet-delivery model, edge-compute budget, in-process broker.	Literature-grounded channel; 112 unit tests.
4 — CARLA integration + ablation	Camera projection, CARLA-bound RSU/CAV, the CAV late-fusion core (time-aware fusion + 3-rule hysteresis + time-to-collision ladder), NHTSA-style HDV profiles, the ablation runner.	First penetration sweep + a cross-map / weather / VRU extension (135 cells, 5 findings).

Carried-forward issues that Sprint 5 targets. The Sprint 4 ablation was scientifically useful but had four realism problems: collision counts were inflated by counting contact-frames rather than incidents; every vehicle (including the notionally-autonomous ones) was made hostile; some vehicles spawned overlapping; and there was no vehicle-to-vehicle link — only roadside-to-vehicle.

3. What Sprint 5 changes

Five aims, each a concrete before/after change.

#	Aim	Before (Sprint 4)	After (Sprint 5)
1	Fix the collision counter	Counted contact-frames with a 1 s per-pair window; stuck/interpenetrating cars kept firing the sensor, inflating counts (e.g. ~650).	**One incident per unordered pair**; on contact both vehicles are **immobilised** (autopilot off, full brake + handbrake, physics frozen). Trustworthy counts; no actor destruction (avoids a Windows crash).
2	Add V2V communication	Only RSU -> CAV (V2I). Vehicles could not share what they saw or intended.	CAVs broadcast **perceived objects** (CPM-style) and **own pose + hard-brake intent** (CAM/DENM-style) to nearby CAVs; receivers fuse objects and pre-brake on a leader's warning. Same channel model.
3	Vehicles spawn inside each other	Forced spawn on raw spawn points; nearby points produced overlapping vehicles.	**try_spawn_actor** (skips blocked spawns) + a **minimum-separation filter** on chosen points.
4	Only HDVs should be hostile	Hostile mix applied to **all** vehicles; collision avoidance disabled for all.	**Role split**: HDVs = hostile mix, avoidance OFF (conflict generators). CAVs = new **AI_REALISTIC** profile (~1% rule-error rate, never ignores vehicles/pedestrians), avoidance ON.
5	Run a new ablation	Single-phase sweep with the issues above.	**Two-phase V2V-isolation** sweep (V2V on vs V2I-only) on the Sprint 4 matrix; results in section 5.

Implementation added one module (`v2xsim/v2v.py`), extended four (`hdv, cav, carla_cav, carla_rsu`), and reworked the runner. **Tests grew 215 -> 241, all passing.**

4. Methodology

Ablation design. The headline experiment is a two-phase comparison that isolates the V2V contribution: every cell is run once with the full V2V system and once V2I-only (`--no-v2v`, CAVs keep RSU messages + their own sensor but no CAV-to-CAV link). Differences are therefore attributable to V2V alone.

- **Matrix:** 3 maps (Town05 in-distribution, Town10HD_Opt held-out, Town01) x 3 weather x 3 penetration (0 / 0.5 / 1.0) x 5 seeds.
- **Population:** 60 vehicles + 60 pedestrians; 3 RSUs; 120 s of simulation per cell at max fidelity (20 Hz physics, 10 Hz broadcast).
- **Metrics:** collisions (total and CAV-involved, counted as distinct incidents), braking actions, phantom-brake suppression, cooperative brakes, V2V throughput, channel-busy ratio.
- **Robustness:** one OS process per cell (crash isolation), resumable, per-cell timeout, wall-clock deadline, and a CARLA-server watchdog.

Comparability caveat. Because Sprint 4 counted contact-frames and Sprint 5 counts incidents, the *absolute* collision numbers are not comparable across sprints. The valid comparisons are **trends** (shape vs penetration) and the **V2V-on vs V2V-off** contrast within Sprint 5.

5. Results

5.1 Town05 — complete V2V on/off isolation (mean over 3 weather x 5 seeds, n=15 per cell). Collisions shown as total / CAV-involved.

Penetration	V2I-only coll.	+V2V coll.	V2I hard-brakes	+V2V hard-brakes
0.0 (all HDV)	27.5 / 0.0	27.5 / 0.0	0	0
0.5 (mixed)	14.7 / 7.3	15.5 / 7.3	2,782	13,892
1.0 (all CAV)	2.0 / 2.0	3.2 / 3.2	6,543	45,651

5.2 Town10HD_Opt — cross-map generalisation (V2V phase; held-out map).

Penetration	+V2V coll. (total / CAV)	hard-brakes	n
0.0	42.0 / 0.0	0	10
0.5	26.3 / 15.0	13,756	10
1.0	8.5 / 8.5	45,398	4 (others timed out)

5.3 V2V on/off across maps at p=0.5 (the penetration where the two phases overlap on both maps).

Map	V2I-only (total / CAV)	+V2V (total / CAV)	Change
Town05 (easy, in-dist, n=15)	14.7 / 7.3	15.5 / 7.3	~0 (no benefit)
Town10HD_Opt (hard, held-out, n~10)	32.2 / 20.8	**26.3 / 15.0**	**-18% / -28%**

Findings

Finding 1 — penetration is the dominant safety lever, and it generalises. Replacing hostile HDVs with careful CAVs collapses collisions monotonically on both the in-distribution map (27.5 -> 14.7 -> 2.0) and the held-out Town10 (42 -> 26 -> 8.5). The Sprint 5 role split is the headline win and it is not a single-map artefact.

Finding 2 — V2V's marginal value over roadside-only perception depends on scenario difficulty. On easy Town05 the RSU coverage already catches most conflicts, so V2V adds only braking churn (~5x more hard-brakes) with no collision benefit. On the harder, denser held-out Town10, V2V **cuts collisions 18% and CAV-involved collisions 28%** at p=0.5 — the extra cooperative perception pays off where there is more to catch.

Finding 3 — reactive cooperative braking has a cost. V2V multiplies hard-braking 3–5x, and at full penetration on Town05 it is slightly worse than V2I-only (2.0 -> 3.2): a saturation / string-instability effect of purely reactive braking. This points to **coordinated** (rather than independently reactive) longitudinal control as the natural next step — consistent with the Sprint 4 'saturation and rebound' observation, now cleanly isolated.

6. Performance characterisation

During the run the GPU and CPU sat at low utilisation while the simulator window stuttered — the classic signature of a workload that is **waiting, not computing**.

- **Synchronous-mode ping-pong:** client and server alternate (one ticks while the other is idle), so neither is ever saturated.
- **Per-actor RPC flurry:** the runner originally made one CARLA remote call per actor per tick (poses, sensor snapshot, control). At full penetration with 120 actors that is ~500 round-trips/tick — ~150 ms of pure wire latency per frame.
- **O(N^2) V2V message storm:** at p=1.0, 60 CAVs broadcasting to each other generate ~500,000 messages per 30 s of simulation, all decoded and fused by a single-threaded Python broker.

Fix and outcome. All reads were batched into a single per-tick world snapshot and all vehicle controls into one batched call. This gave a ~1.4x speedup and stopped the heaviest cells timing out, but it exposed the V2V message volume as the real ceiling at high penetration. That ceiling is **itself a result**: cooperative-perception load scales quadratically with connected-vehicle density — directly relevant to the project's edge-compute and channel-congestion theme.

Practical consequence. A full 3-map x both-phase sweep at maximum fidelity does not fit an overnight window. Sprint 5 therefore prioritised and completed the clean Town05 V2V on/off comparison plus the Town10 cross-map generalisation; the remaining cells (all of Town01, the Town10 V2I pair, and reliable Town10 p=1.0) are documented follow-ups that need either a faster timestep setting or a multi-day budget.

7. Conclusions and future work

- **The realism fixes worked:** collision counts are now trustworthy, roles are physically meaningful (hostile humans vs careful autonomy), and spawns are clean.
- **Safety scales with connected-vehicle penetration**, and the effect generalises to a held-out map.
- **V2V helps where the scenario is hard enough to need it**, at a braking-churn cost — motivating coordinated longitudinal control.
- **Cooperative-perception load is quadratic** in CAV density, a concrete edge-compute / channel finding.

Future work: coordinated (non-reactive) cooperative braking; complete the cross-map V2V-isolation (Town01 + Town10 V2I) at a faster timestep; per-detection confidence weighting to handle lower detector quality on out-of-distribution maps.

8. Appendix

New / changed code (Sprint 5)

File	Role
v2xsim/v2v.py (new)	V2V message: perceived objects + pose + hard-brake intent
v2xsim/hdv.py	AI_REALISTIC profile; role-aware avoidance disable
v2xsim/cav.py	Cooperative brake-warning ingestion + reaction
v2xsim/carla_cav.py	V2V send/receive; snapshot ego-state path
scripts/40_ablation_run.py	Incident counting + immobilise; role split; V2V loop; spawn fix; batching
scripts/42_..._subprocess.py	--no-v2v, --max-hours deadline, server watchdog

New runner knobs

Flag	Default	Purpose
--no-v2v	off	V2I-only arm (isolate V2V)
--v2v-range	150 m	CAV-to-CAV broadcast range
--cpm-period-ms	100 (10 Hz)	Broadcast cadence; 200 ms ~1.7x faster
--dt	0.05 (20 Hz)	Physics timestep; 0.1 ~2.3x faster, coarser
--cav-attentive	off	CAVs flawless instead of AI_REALISTIC

Detailed change log and raw cell data: `SPRINT5_RESULTS.md`, `out/ablation_sprint5_full/{v2v,noV2V}/summary*.csv`.